

SEO & Schema Parity — Target State Design

AEM 6.4 to AEM as a Cloud Service (Edge Delivery
Services) Migration

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Adobe Professional Services – Edge Delivery Services



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SEO & Schema Parity — Target State Design

1. Executive Summary

This section defines the SEO parity strategy for migrating thermofisher.com from AEM 6.4 to AEM as a Cloud Service with Edge Delivery Services (Universal Editor). It ensures zero regression in search visibility, structured data richness, and metadata completeness during and after migration.

2. Schema.org Structured Data Inventory (Source State)

2.1 Schema Types Identified on Current Site

Schema Type	Page Type	Implementation	Example Page
WebSite + SearchAction	Homepage	JSON-LD	/us/en/home.html
Organization	Homepage	JSON-LD	/us/en/home.html
ItemList > BlogPosting	Homepage (blog feed)	JSON-LD	/us/en/home.html
BreadcrumbList	Category/L2+ pages	HTML markup (visible breadcrumb)	/us/en/home/life-science/pcr.html
Product	Product detail pages	JSON-LD (expected)	/order/catalog/product/*
FAQPage	Support/FAQ pages	JSON-LD (expected)	FAQ/support pages

2.2 Detailed Schema Definitions

WebSite (Homepage)

```
{
  "@context": "http://schema.org",
  "@type": "WebSite",
  "url": "https://www.thermofisher.com/",
  "name": "Thermo Fisher Scientific",
  "potentialAction": {
    "@type": "SearchAction",
    "target": "https://www.thermofisher.com/search/results?query={search_term_string}&resultPage=1&resultsPerPage=15&autocomplete=",
    "query-input": "required name=search_term_string"
  }
}
```

Organization (Homepage)

```
{
  "@context": "http://schema.org",
  "@type": "Organization",
  "name": "Thermo Fisher Scientific",
  "url": "https://www.thermofisher.com/",
  "sameAs": [
```

```
"https://twitter.com/thermofisher",
"https://www.facebook.com/thermofisher",
"https://plus.google.com/+thermofisher",
"https://www.linkedin.com/company/thermo-fisher-scientific",
"https://www.youtube.com/channel/UCfUs2fCDhx07fkszsJ0j0cA"
],
"description": "Thermo Fisher Scientific is an American multinational, biotechnology
product development company...",
"logo": "https://www.thermofisher.com/logo.jpg",
"telephone": "+1-800-711-2088",
"address": {
  "@type": "PostalAddress",
  "streetAddress": "5823 Newton Drive",
  "addressLocality": "Carlsbad",
  "addressRegion": "CA",
  "postalCode": "92008"
}
}
```

ItemList > BlogPosting (Homepage)

```
{
  "@context": "https://schema.org",
  "@type": "ItemList",
  "itemListElement": [
    {
      "@type": "ListItem",
      "position": 1,
      "item": {
        "@type": "BlogPosting",
        "headline": "...",
        "url": "https://www.thermofisher.com/blog/...",
        "datePublished": "2026-05-04T00:00:00+00:00",
        "image": { "@type": "ImageObject", "url": "..." },
        "author": { "@type": "Person", "name": "..." }
      }
    }
  ]
}
```

3. Metadata Inventory (Source State)

3.1 Standard Meta Tags

Meta Tag	Present On	Example Value	Required in Target
<title>	All pages	Polymerase Chain Reaction (PCR) \\ Thermo Fisher Scientific - US	Yes
meta[name="description"]	All pages	Page-specific description (150–200 chars)	Yes
meta[name="robots"]	All pages	index, follow	Yes

Meta Tag	Present On	Example Value	Required in Target
meta[name="viewport"]	All pages	width=device-width, initial-scale=1.0...	Yes (EDS default)
meta[name="DC.title"]	All pages	Dublin Core title	Yes (legacy parity)

3.2 Open Graph Protocol

Property	Present On	Example	Required in Target
og:title	All pages	Matches <title>	Yes
og:type	All pages	website	Yes
og:url	All pages	Canonical URL	Yes
og:description	All pages	Matches meta description	Yes
og:locale	All pages	en_US	Yes
og:image	All pages	Brand image or page-specific	Yes

3.3 Search Engine Verification Tags

Tag	Purpose	Value
baidu-site-verification	Baidu Webmaster Tools	codeva-AlvwygfY0A
sogou_site_verification	Sogou Search	wdyDZ0DQ8G

3.4 Link Elements

Element	Purpose	Count/Scope
<link rel="canonical">	Canonical URL declaration	All pages
<link rel="alternate" hreflang="...">	International targeting	39 locales per page
<link rel="dns-prefetch">	Performance hint	downloads.thermofisher.com, assets.adobedtm.com, etc.
<link rel="preconnect">	Performance hint	assets.adobedtm.com, edge.adobedc.net, dm- images.thermofisher.com

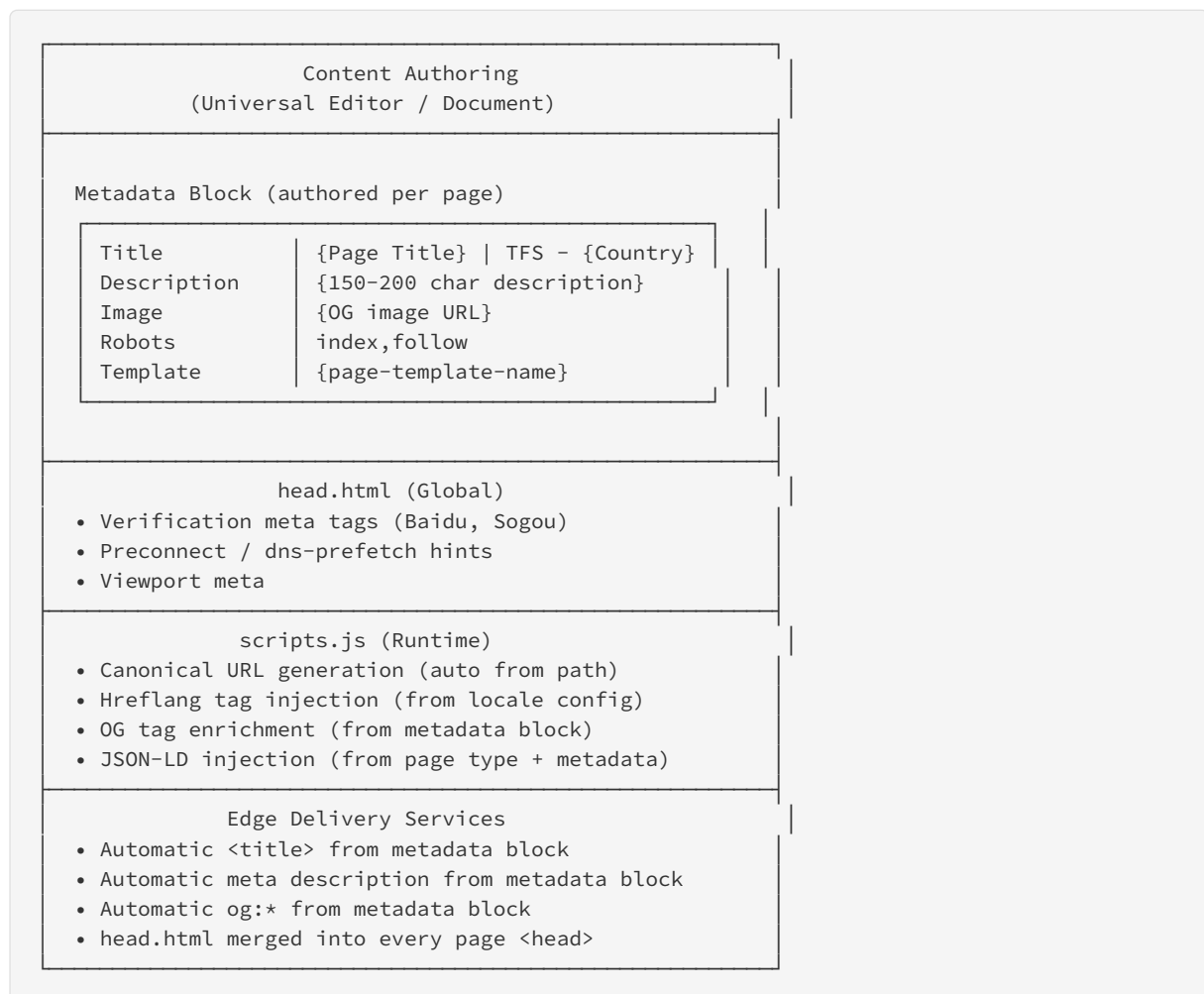
4. Hreflang Coverage (39 Locales)

Locale	Country/Language	URL Pattern
en-us	United States / English	thermofisher.com/us/en/
es-es	Spain / Spanish	thermofisher.com/es/es/
fr-fr	France / French	thermofisher.com/fr/fr/
de-de	Germany / German	thermofisher.com/de/de/
ja-jp	Japan / Japanese	thermofisher.com/jp/ja/

Locale	Country/Language	URL Pattern
ko-kr	Korea / Korean	thermofisher.com/kr/ko/
zh-cn	China / Chinese	thermofisher.cn/cn/zh/
zh-tw	Taiwan / Chinese	thermofisher.com/tw/zt/
pt-br	Brazil / Portuguese	thermofisher.com/br/pt/
ru-ru	Russia / Russian	thermofisher.com/ru/ru/
...	(29 additional locales)	Same pattern

5. Target State Generation Strategy (EDS / Universal Editor)

5.1 Metadata Generation Architecture



5.2 Structured Data Generation Strategy

Schema Type	Generation Method	Trigger
WebSite + SearchAction	Static JSON-LD in head.html (homepage only) or conditional injection in scripts.js	Homepage template
Organization	Static JSON-LD in head.html (homepage only) or global footer script	Homepage template
BreadcrumbList	Auto-generated by breadcrumb block from URL path + page titles	Breadcrumb block present on page
ItemList > BlogPosting	Dynamic JSON-LD generated from blog feed data	Homepage blog section
Product	JSON-LD generated from product data API/metadata	Product detail template
FAQPage	Auto-generated from FAQ/accordion block content	FAQ block present on page

5.3 Canonical URL Strategy

Scenario	Rule
Standard pages	Auto-generate from window.location path (strip query params, trailing slash normalization)
Paginated content	Add ?page=N to canonical
Locale variants	Canonical points to self; hreflang handles alternates
Duplicate content	Author specifies canonical override in metadata block

5.4 Hreflang Generation Strategy

Approach: Configuration-driven injection at runtime via a locale mapping spreadsheet.

/locale-config.json (authored in spreadsheet)

hreflang	country-code	base-url
en-us	us	https://www.thermofisher.com/us/en
fr-fr	fr	https://www.thermofisher.com/fr/fr
zh-cn	cn	https://www.thermofisher.cn/cn/zh
...	...	...

Runtime logic (scripts.js or delayed.js): 1. Fetch locale config on page load 2. For each locale, construct the alternate URL by replacing country/language segments 3. Inject <link rel="alternate" hreflang="..."> tags into <head> 4. Include x-default pointing to the US English variant

5.5 Metadata Block Mapping (Source → Target)

Source (AEM 6.4 JCR)	Target (EDS Metadata Block)	Notes
jcr:title / pageTitle	Metadata → Title	Auto-appended with \ Thermo Fisher Scientific - {Country}

Source (AEM 6.4 JCR)	Target (EDS Metadata Block)	Notes
jcr:description	Metadata → Description	150–200 chars, keyword-rich
cq:tags	Metadata → Tags	Taxonomy mapping required
navTitle	Used for breadcrumb trail	Fetches via page title resolution
hideInNav	Metadata → Hide in Nav	Boolean
noIndex	Metadata → Robots	Outputs noindex, follow
og:image (custom property)	Metadata → Image	Default fallback to brand OG image
canonicalUrl (if override)	Metadata → Canonical	Only when different from self

6. Validation & Regression Checks

6.1 Automated Pre-Migration Baseline

Check	Tool	Frequency	Output
Crawl full site metadata	Screaming Frog / Sitebulb	Once (before migration)	Baseline CSV of all titles, descriptions, canonicals, robots
Schema.org validation	Google Rich Results Test API	Once per template type	Baseline structured data per page type
Hreflang audit	Screaming Frog hreflang report	Once (before migration)	Full locale mapping verification
Core Web Vitals baseline	PageSpeed Insights / CrUX	Once (before migration)	LCP, CLS, INP baselines

6.2 Post-Migration Regression Test Suite

#	Test	Expected Result	Pass Criteria	Tool
1	Title tag parity	Every migrated page has matching <title>	100% match or approved deviation	Crawl comparison
2	Meta description parity	Every migrated page has description present	100% present, >95% match	Crawl comparison
3	Canonical URL correctness	Self-referencing canonical on all pages	0 broken/missing canonicals	Crawl audit
4	Robots directives	index, follow or author-specified noindex	0 unintended noindex	Crawl comparison
5	OG tag completeness	og:title, og:description, og:url, og:image on all pages	100% presence	Crawl audit
6	Hreflang count	39 hreflang tags per page (matching source)	Count match ±0	Crawl audit
7	Hreflang reciprocity	All hreflang links are reciprocal	0 orphaned hreflangs	Screaming Frog
8	JSON-LD WebSite schema	Present on homepage with SearchAction	Valid per Rich Results Test	Google Rich Results Test

#	Test	Expected Result	Pass Criteria	Tool
9	JSON-LD Organization schema	Present on homepage with sameAs, logo, address	Valid per Rich Results Test	Google Rich Results Test
10	JSON-LD Breadcrumb-List	Present on L2+ category pages	Valid hierarchical structure	Google Rich Results Test
11	JSON-LD BlogPosting	Present on homepage (or blog pages)	Valid per Rich Results Test	Google Rich Results Test
12	Breadcrumb path accuracy	Breadcrumb matches URL hierarchy	Path segments resolve to correct titles	Visual + crawl check
13	Sitemap.xml generation	Sitemap generated from EDS page index	All published pages included, valid XML	XML validator + compare
14	robots.txt correctness	Correct disallow/allow rules	Matches source robots.txt directives	Manual review
15	HTTP status codes	Migrated URLs return 200; old URLs 301 to new	0 unexpected 404s	Crawl audit
16	301 redirect mapping	All old URLs redirect to correct new URLs	100% coverage for indexed pages	Redirect test script
17	Page load performance	CWV scores ≥90 (PageSpeed)	No regression from baseline	PageSpeed Insights
18	Search Console indexing	No indexing errors post-migration	0 new errors within 30 days	Google Search Console

6.3 Continuous Monitoring (Post-Go-Live)

Metric	Monitoring Tool	Alert Threshold
Indexed pages count	Google Search Console	Drop >5% week-over-week
Rich results count	Google Search Console	Any decrease
Crawl errors	Google Search Console	New errors >0
Core Web Vitals	CrUX Dashboard	Any metric failing threshold
Organic traffic	Adobe Analytics / GA4	Drop >10% vs. pre-migration baseline
Ranking positions	SEMrush / Ahrefs	Average position increase >3

6.4 Migration Testing Phases

Phase 1: Template Validation (Pre-Migration)

- Validate metadata block rendering for each template type
- Validate JSON-LD output per schema type
- Validate hreflang injection logic
- Run Google Rich Results Test on preview URLs

Phase 2: Batch Migration QA (During Migration)

- Compare crawl of migrated batch vs. source baseline
- Flag any missing titles, descriptions, canonicals
- Validate redirect mapping for batch
- Spot-check structured data on sample pages per template

Phase 3: Full-Site Regression (Post-Migration, Pre-Go-Live)

- Full crawl comparison (source vs. target)
- Hreflang reciprocity audit
- Redirect chain validation (no chains >1 hop)
- Performance regression test (LCP, CLS, INP)
- Stakeholder sign-off on SEO dashboard

Phase 4: Post-Go-Live Monitoring (30/60/90 days)

- Daily: Crawl errors, indexing status
- Weekly: Organic traffic, ranking positions
- Monthly: Rich results, Core Web Vitals
- Quarterly: Full SEO health audit

7. Risk Mitigations

Risk	Mitigation
Hreflang tags missing on EDS pages	Runtime injection validated via automated tests; fallback to <code>head.html</code> static tags
JSON-LD schema breaks on content changes	Schema templates tied to page templates; content changes don't affect schema structure
Redirect loops or chains	Redirect map validated pre-go-live; monitoring for chains >1 hop
Loss of indexed pages	Phased migration with overlap period; Search Console monitoring; rapid rollback plan
Duplicate content (old + new coexisting)	Canonical tags always point to target; robots.txt blocks old paths after cutover
Metadata author errors	Validation rules in Universal Editor; required fields enforced; preview shows rendered SEO

8. Deliverables Checklist

#	Deliverable	Owner	Status
1	Pre-migration crawl baseline (titles, descriptions, canonicals, schemas)	SEO Lead	Pending
2	Hreflang locale configuration spreadsheet	SEO Lead + i18n	Pending
3	Redirect mapping (old URLs → new URLs)	SEO Lead + Dev	Pending
4	JSON-LD schema templates (per page type)	Dev	Pending
5	head.html with global SEO elements	Dev	Pending
6	Metadata block authoring guide for content authors	Content Lead	Pending

#	Deliverable	Owner	Status
7	Automated regression test scripts	QA/Dev	Pending
8	Post-go-live monitoring dashboard	SEO Lead	Pending
9	Rollback playbook (if SEO regression detected)	DevOps	Pending